SECTION A

1. (a) An inclined (θ=30°) simple beam is subject to a uniformly distributed load q along AB as shown in Figure 1(a).

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

   (ii) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tensile** side. **(5 marks)**

   (b) A frame is subject to a horizontal force F applied at point D, as illustrated in Figure 1(b). A is a pin support and B is a roller support.

   (i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

   (ii) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tensile** side. **(5 marks)**

2. (a) As shown in Figure 2(a), a bar AB consisting of 2 elements, each of which is homogeneous and of uniform cross section, is subjected to an axial load P applied at point C. E_1 and E_2 denote the modulus of elasticity of the material used in the elements. The area of cross section is A.

   (i) Draw the axial force diagram (AFD) with the sign labels and pertinent values. Note that positive axial force is **tensile**. **(3 marks)**

   (ii) Find the axial displacement at point A and C. **(5 marks)**

   (b) An axial load P is applied at point C of a fixed-end bar ACB (Figure 2(b)), which is also made of 2 elements with homogeneous material and uniform cross section in each. E_1 and E_2 denote the modulus of elasticity of the material used in the elements AC and CB, respectively. The area of cross section is A.

   (i) if E_1=E_2=E, draw the axial force diagram (AFD) with the sign labels and pertinent values and find the axial displacement at point C. Express the axial displacement in terms of P, L, E and A. Note that positive axial force is **tensile**. **(6 marks)**

   (ii) if $E_1 \neq E_2$, draw the axial force diagram (AFD) with the sign labels and pertinent values and find the axial displacement at point C. Express the axial displacement in terms of P, L, E_1, E_2 and A. Note that positive axial force is **tensile**. **(6 marks)**

   Hint: For the fixed-end bar, the displacement at point A is zero.

3. (a) A vertical column AB of square cross section is loaded by a horizontal force P at the free end (Figure 3(a)). The side of square cross section is a, as shown in figure 3(c). The modulus E = 200GPa, length L = 0.5m, a=100mm.

(i) Draw the bending moment diagram (BMD) with pertinent values. Note that bending moment should be drawn on the **tensile** side. **(2 marks)**

(ii) Find the deflection at point B and the maximum positive stress as well as its position.

(6 marks)

(b) Two vertical columns AB and CD are connected by truss element BC, as shown in Figure 3(b). A concentrated load (P = 100kN) is applied at point B horizontally. The material properties and geometry of two columns are same. The cross section of all the structural elements is square with a side length a = 100mm (see figure 3(c)). The modulus of columns E = 200GPa, length L = 0.5m.

(i) If EA=0, Draw the bending moment diagram (BMD) with pertinent values and calculate the deflection at point B using the given values. Note that bending moment should be drawn on the **tensile** side. **(6 marks)**

(ii) If $EA \rightarrow \infty$, Draw the bending moment diagram (BMD) with pertinent values and calculate the deflection at point B using the given values. Note that bending moment should be drawn on the **tensile** side. **(6 marks)**

FIGURES

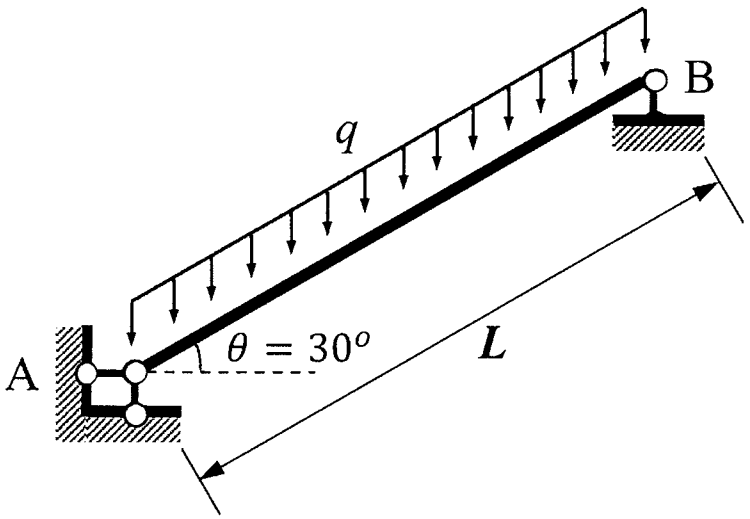


Figure 1(a) Beam

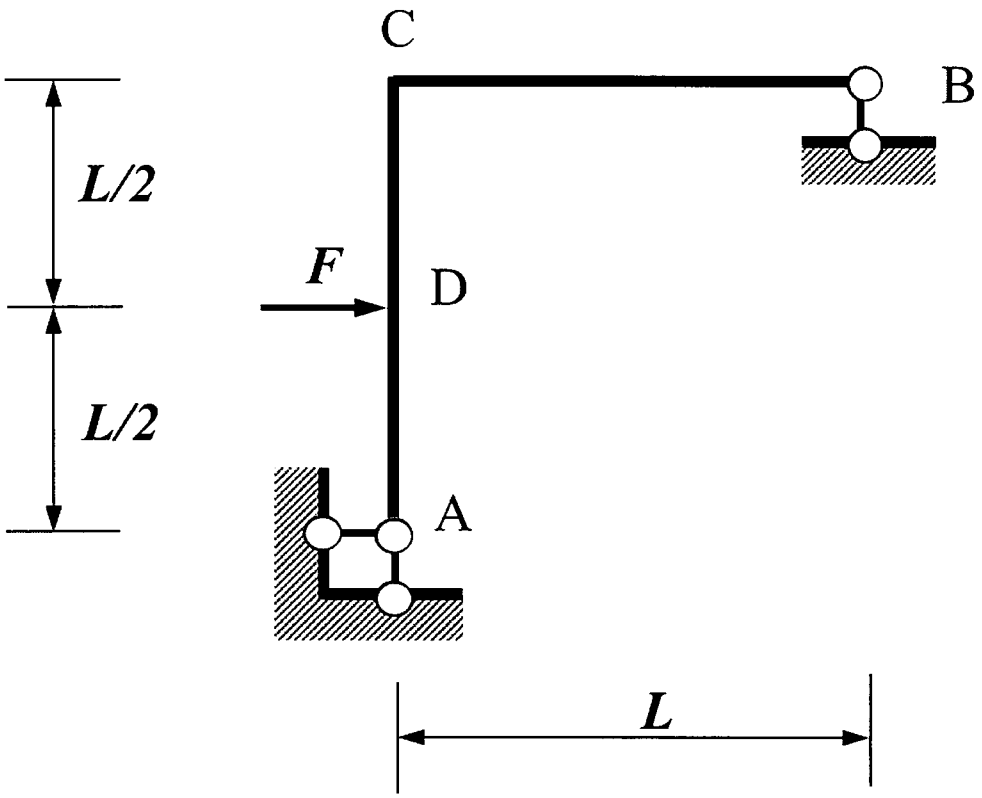


Figure 1(b) Frame

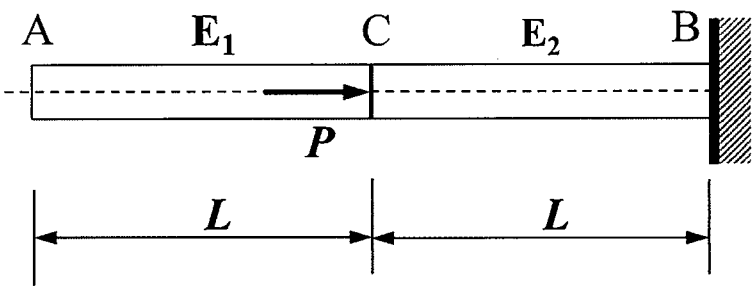


Figure 2(a) Bar fixed at B

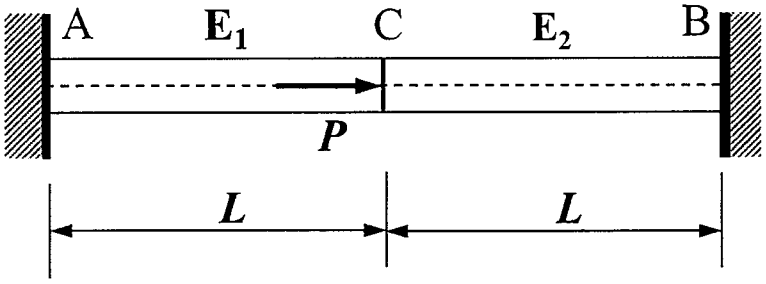


Figure 2(b) Fixed-end Bar (Fixed at both A and B)

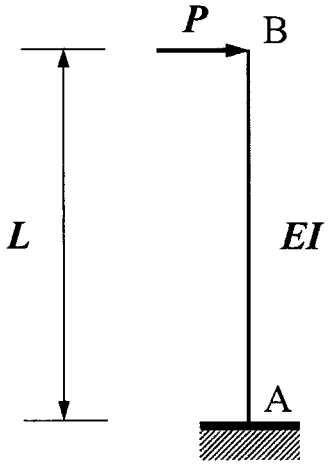


Figure 3(a) Single column AB

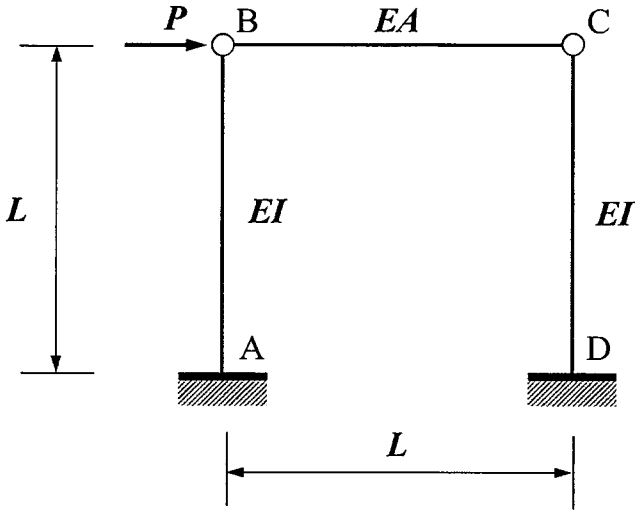


Figure 3(b) Two columns connected by truss element

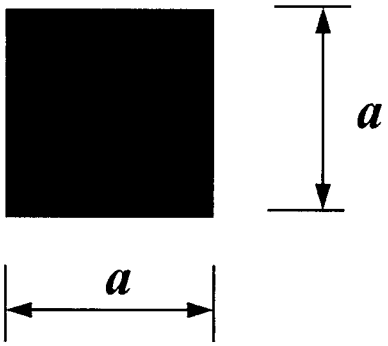


Figure 3(c) Cross section

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